FSOC Data Rate calculations.

Eng. Carlos German Carreño Romano

Digital Communication

Baud Rate

Bandwidth

Modulation

Spectral Efficiency

Capacity

$$C = B \cdot \log_2\left(1 + \frac{S}{N}\right)$$

Signal-to-Noise Ratio

$$\mathrm{SNR}_0 = \frac{i_s}{\sigma_N} = \sqrt{\frac{\eta P_S}{2h\nu B}} \quad \rightarrow \quad 10 \cdot \log(\mathrm{SNR}_0) = 74.9 \text{ dB}$$

$$i_S = \frac{\eta e P_s}{h\nu} = \frac{0.85 \cdot e \cdot 10 \text{ mW}}{h \cdot 193.4 \text{ THz}} = 0.01063$$

$$\left\langle i_N^2 \right\rangle = 2eBi_S = 2 \cdot e \cdot 100 \cdot 10^9 \cdot 0.01063 = 3.406 \cdot 10^{-10}$$

Bit Error Rate

OOK: On-Off Keying

QPSK: Quadrature Phase Keying

QAM: Quadrature Amplitude Modulation

Data Rate

Photodetectors

Noise

APD

PIN

Laser transmitter

Amplitude modulation

Phase modulation

Polarization modulation

Laser beam propagation

Free-space

$$\Theta_0 = 1 - \frac{L}{F_0}$$

$$\Lambda_0 = \frac{2L}{kW_0^2}$$

$$W = W_0\sqrt{\Theta_0^2 + \Lambda_0^2}$$

$$I(0, L) = I_0 \cdot \frac{W_0^2}{W^2} \left[\frac{\mathrm{W}}{\mathrm{m}^2}\right]$$

$$P_{R_x} = P_{T_x} \cdot \left(1 - e^{\frac{-2r^2}{W^2}}\right) [\mathrm{W}]$$

$$I_{R_x} = 4\frac{P_{R_x}}{\pi D_{R_x}^2} \left[\frac{\mathrm{W}}{\mathrm{m}^2}\right]$$

Random media

$$\langle \mathrm{SNR} \rangle = \frac{\mathrm{SNR}_0}{\sqrt{\left(\frac{P_{S0}}{\langle P_S \rangle}\right) + \sigma_I^2(D_G)\mathrm{SNR}_0^2}}$$

$$\langle \mathrm{SNR} \rangle \cong \frac{1}{\sigma_1(D_G)}, \quad \mathrm{SNR}_0 \to \infty$$

Fading statistics

$$Pr_{\mathrm{fade}} = 1 - \frac{1}{2}\int_0^\infty (u)\mathrm{erfc}\left(\frac{\mathrm{TNR} - \langle \mathrm{SNR} \rangle\, u}{\sqrt{2}}\right) du$$

$$Pr(E) = \langle \mathrm{BER} \rangle = \frac{1}{2}\int_0^\infty p_f(u)\mathrm{erfc}\left(\frac{\langle \mathrm{SNR} \rangle\, u}{2\sqrt{2}}\right) du$$